

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Transformations of Graphs

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

TRANSFORMATIONS OF GRAPHS · CH₃ · L₉

ENG. ESLAM AHMED



The Map

This lesson collapses Transformations of Graphs into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Shift up or down

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Shift left or right

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Stretch vertically

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Reflect or scale horizontally

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 38 website questions, 38 checked solutions, 3 local PDFs read.

01 Shift up or down

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations of Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$a = 2.5$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q22

Read the graph

$$a = 2.5$$

Use trigonometry

$$x + b = 360$$

Finish: $a = 2.5$, $b = 300$; $(2, 5)$ and $(4, -2)$

FORMULA BANK

$f(x) + a$ shifts up, $f(x - a)$ shifts right, $af(x)$ stretches vertically, $f(ax)$ changes horizontally.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

22 The graph of $y = a \cos(x + b)$ degrees for $0 < x < 360$ is drawn on the grid. $y \times 360$
 300 240 180 120 60 O -3 -2 -1 1 2 3 (a) Find the value of a and the value of b . $a = b = (2)$
 Another curve C has equation $y = f(x)$ The coordinates of the minimum point of C are
 (4, 5) (b) Write down the coordinates of the minimum point of the curve with equation
 (i) $y = f(2x)$ (,) (ii) $y = f(x) - 7$ (,) (2)

YOUR SOLUTION

02 Shift left or right

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations of Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$a = 2.5$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q22

Read the graph

$$a = 2.5$$

Use trigonometry

$$x + b = 360$$

Finish: $a = 2.5$, $b = 300$; $(2, 5)$ and $(4, -2)$

FORMULA BANK

$f(x) + a$ shifts up, $f(x - a)$ shifts right, $af(x)$ stretches vertically, $f(ax)$ changes horizontally.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

22 The graph of $y = a \cos(x + b)$ degrees for $0 < x < 360$ is drawn on the grid. $y \times 360$
 300 240 180 120 60 O -3 -2 -1 1 2 3 (a) Find the value of a and the value of b . $a = b = (2)$
 Another curve C has equation $y = f(x)$ The coordinates of the minimum point of C are
 (4, 5) (b) Write down the coordinates of the minimum point of the curve with equation
 (i) $y = f(2x)$ (,) (ii) $y = f(x) - 7$ (,) (2)

YOUR SOLUTION

03 Stretch vertically

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations of Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = f(x + 4)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q21

The maximum point of is

This is a reasoning step: write one clear sentence, then continue the calculation.

Calculate value

$$y = f(x + 4)$$

Finish: $(0, 6)$, $(2, 6)$, and $y = (x - 4)^2 + 3(x - 4) + 10$

FORMULA BANK

$f(x) + a$ shifts up, $f(x - a)$ shifts right, $af(x)$ stretches vertically, $f(ax)$ changes horizontally.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

21 The diagram shows a sketch of part of the curve with equation $y = f(x)$. The point $(4, 6)$ is a maximum point on this curve. The coordinates of this maximum point are $(4, 6)$.

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 4)$ (,) (ii) $y = f(2x)$ (,)

(2) The equation of a curve C is $y = x^2 + 3x + 4$. The curve C is transformed to curve S under the translation $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$.

(b) Find an equation of curve S. You do not need to simplify the equation. (2)

YOUR SOLUTION

04 Reflect or scale horizontally

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations of Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = f(x + 4)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q21

The maximum point of is

This is a reasoning step: write one clear sentence, then continue the calculation.

Calculate value

$$y = f(x + 4)$$

Finish: $(0, 6)$, $(2, 6)$, and $y = (x - 4)^2 + 3(x - 4) + 10$

FORMULA BANK

$f(x) + a$ shifts up, $f(x - a)$ shifts right, $af(x)$ stretches vertically, $f(ax)$ changes horizontally.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations of Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$a = 2.5$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q22

Read the graph

$$a = 2.5$$

Use trigonometry

$$x + b = 360$$

Finish: $a = 2.5$, $b = 300$; $(2, 5)$ and $(4, -2)$

FORMULA BANK

$f(x) + a$ shifts up, $f(x - a)$ shifts right, $af(x)$ stretches vertically, $f(ax)$ changes horizontally.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Shift up or down
"work out", "show", "hence", a familiar exam phrase	Shift left or right
"work out", "show", "hence", a familiar exam phrase	Stretch vertically
"work out", "show", "hence", a familiar exam phrase	Reflect or scale horizontally
"work out", "show", "hence", a familiar exam phrase	Read the graph

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.